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In-vitro regeneration in long-day garlic (*Allium sativum*)

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ABSTRACT

The cloves of garlic (*Allium sativum* L.) cultivar Mukteshwar Local, after disinfection, were sprouted under *in-vitro* condition (85.00 % sprouting) on MS medium containing 2 mg/l GA^3. The cloves were soaked in 100 ppm GA^3 solution for 24 hr. Maximum number of shoot buds (7.0) was recorded in treatment containing MS + 4.0 ppm BAP + 1.0 ppm NAA. Among different treatments for shoot multiplication, treatment MS+5.0 ppm BAP+ 1.0 ppm KIN + 0.5 ppm GA^3 was the best. The number of shoots also varied with the level of sugar, light intensity and, light and dark cycle on shoot multiplication. They multiplied best in 30 g/l sucrose, 5000 lux light intensity and 16:8 hr of light and dark cycle. The developed protocol may be used for mass multiplication of garlic cultivars as well as regeneration of genetically transformed cell/ tissue.

Cultivated garlic (*Allium sativum* L.) is a sexually sterile crop and exclusively propagated vegetatively. The propagation rate of garlic in field is approximately 5-10% per year. Thus, it takes many years to build up the required material for commercial cultivation of a new variety. There is no denying in the fact that after traditional method of propagation, micropropagation is next step for rapid multiplication of desired genotype. Therefore, *in-vitro* technique, micropropagation, offers great potential to deliver a large quantity of disease-free and true- to- type, healthy stock in a short time.

MATERIALS AND METHODS

The garlic cultivar, Mukteshwar Local, was taken for study at the Central Institute of Temperate Horticulture (ICAR), Regional Station Mukteshwar, Nainital, Uttarakhand, India. Cloves were taken as a source of explants. These were washed with tween-20 and then surface sterilized with 0.1 % $HgCl_2$ for 2 min. After that explant were washed thoroughly with sterile distilled water. The sterilized explants were soaked in distilled water/various concentration of GA_3 for different time period. Then explants were inoculated on MS basal medium containing 3% sucrose supplemented with or without different concentration of GA_3 for *in-vitro* sprouting of cloves. The medium is gelled with 8 g/l agar and pH was adjusted to 5.8 prior to autoclaving at 121°C for 15 min. Observations were recorded on percentage sprouting and days taken to sprout.

The sprouted cloves were inoculated in MS medium supplemented with different concentration of BAP and NAA. Observations were recorded for number of shoot-bud induction and regeneration frequency. The regenerated shoots were then subcultured onto multiplication medium supplemented with different growth regulators. Observations on average number of shoots per culture and mean shoot length were recorded after four weeks of culture. Effect of sucrose level, light intensity and photoperiod on shoot multiplication were recorded.

The factorial completely randomized design (CRD) with three replications was followed. Fifteen samples were maintained in each replication. Percentage data were subjected to Arc Sin transformation before analysis. The analyses of data were done by using SPSS 13.0 software.

RESULTS AND DISCUSSION

Based on sprouting per cent and days required for sprouting, medium and pre-sterilization treatments

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were standardized. The sprouting per cent was higher (85.00%) on MS medium containing 2 mg/l GA3 (M2) when cloves were soaked in 100 mg/l GA3 solution for 24 hr than on MS medium devoid of GA^3 which had 65% sprouting (Table 1). Days required for sprouting was lower in medium containing 2 mg/l GA3 (M2) for all the cultivars than in medium which was devoid of GA3. Days taken for sprouting were minimum (30.0) in medium MS+ 2 ppm GA3 + soaking of clove for 24 hr in 100 ppm GA^3 solution. With the increase/addition of GA^3 either in medium or in solution for pre-treatment increased sprouting considerably.

The findings indicate that gibberellic acid promotes seed germination/sprouting (Salisbury and Ross, 1995) through one or several possible steps, viz. activation of vegetative growth of the embryo, weakening of a growth constraining endosperm layer surrounding embryos and the mobilization of stored food reserve of the endosperm (Taiz and Zeiger, 2002). Enhanced sprouting by GA^3 treatment might also be due to hydrolysation of reserved food material and activation of enzymes which help in germination (Salisbury and Ross, 1995).

Difference was observed among treatments for number of shoot buds per explant. With the increase in concentration of BAP, number of shoot buds increased significantly but at higher concentration (above 4.0ppm BAP), it decreased. Maximum number of shoot bud (7.0) was recorded in treatment containing MS+4.0 ppm BAP+1.0ppm NAA (T^2) and minimum (1.75) was recorded in medium containing MS+7.0 ppm BAP+1.0ppm NAA (T^5). In MS medium containing 4.0 ppm BAP+1.0 ppm NAA, number of days required for shoot bud induction was low, regeneration frequency was maximum and more number of shoot buds was recorded (Table 2, Fig 1). BAP is reported to be the most responsive cytokinin in shoot bud induction in garlic (Haque *et al.* (2003). Cytokinin such as BAP is

Table: 1 Effect of different treatments on sprouting of garlic clove under *in-vitro* conditions

Treatment	Treatment details	Sprouting (%)	Days taken to sprouting
T_1	MS + soaking of clove for 24 hr in distilled water	50.00 (45.00)^{a}	48.22^{d}
T_2	MS + soaking of clove for 24 hr in 50 ppm GA_3 solution	62.50 (52.24)^{ab}	42.33^{bc}
T_3	MS + soaking of clove for 24 hr in 100 ppm GA_3 solution	65.00 (53.73)^{ab}	37.45^{b}
T_4	MS+ 2 ppm GA_3 + soaking of clove for 24 hr in distilled water	58.00 (49.60)^{ab}	45.32^{cd}
T_5	MS+ 2 ppm GA_3 + soaking of clove for 24 hr in 50 ppm GA_3 solution	68.00 (55.55)^{b}	38.51^{b}
T_6	MS+ 2 ppm GA_3 + soaking of clove for 24 hr in 100 ppm GA_3 solution	85.00 (67.21)^{c}	30.24^{a}
		SEm± : 2.68	1.68
		SEd : 3.79	2.37
		$CD_{0.05}$: 8.26	5.17

Data within parentheses are ArcSign transformed value. Means for groups in homogeneous subsets are displayed with same alphabets i.e. treatments denoted with same alphabets are not significant at 5% probability using DMRT.

Table: 2 Effect of different treatments on *in vitro* shoot bud induction on clove explant of garlic

Treatment	Treatment details	Number of shoot buds per explant	Number of days taken for shoot bud induction
T_1	MS+3.0 ppm BAP+1.0ppm NAA	2.00^{a}	58.25^{cd}
T_2	MS+4.0 ppm BAP+1.0ppm NAA	7.00^{d}	45.20^{a}
T_3	MS+5.0 ppm BAP+1.0ppm NAA	6.00^{c}	50.15^{ab}
T_4	MS+6.0 ppm BAP+1.0ppm NAA	4.50^{b}	55.33^{bc}
T_5	MS+7.0 ppm BAP+1.0ppm NAA	1.75^{a}	62.25^{d}
		SEm± : 0.19	2.05
		SEd : 0.26	2.90
		$CD_{0.05}$: 0.58	6.44

Means for groups in homogeneous subsets are displayed with same alphabets, i.e. treatments denoted with same alphabets are not significant at 5% probability using DMRT.

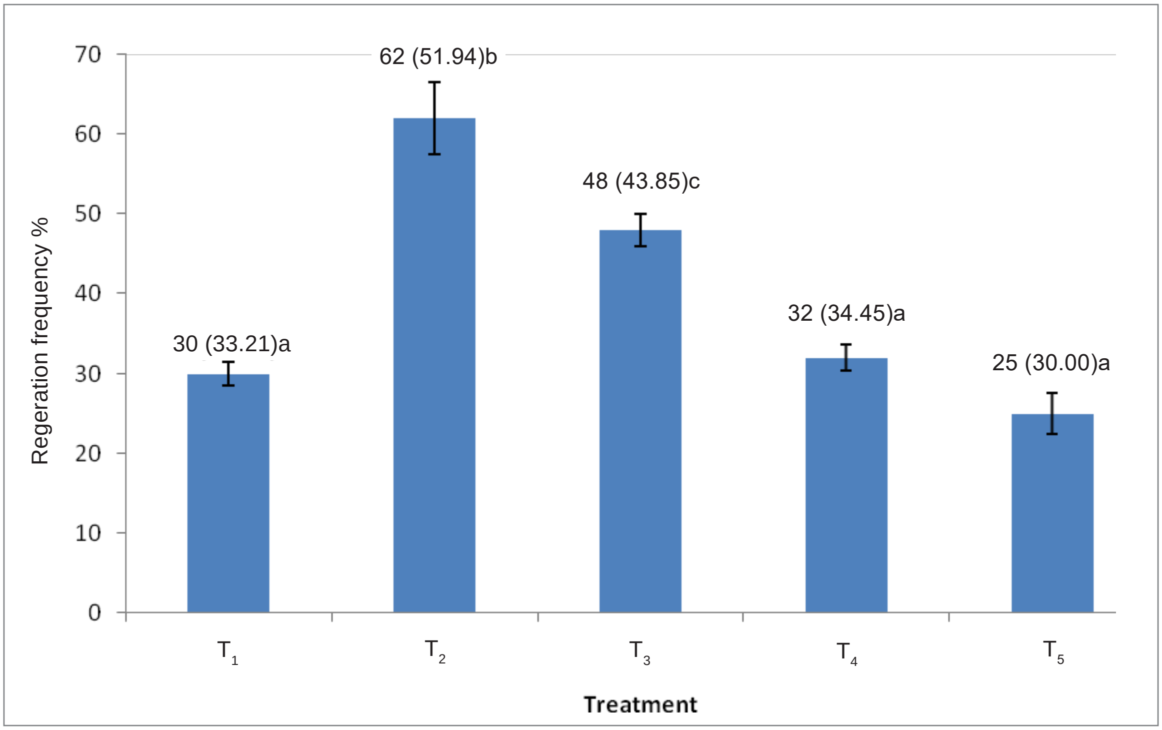

Fig 1. Regeneration frequency of explant in different treatments (T_1: MS+3.0 ppm BAP+1.0ppm NAA; T_2: MS+4.0 ppm BAP+1.0ppm NAA; T_3: MS+5.0 ppm BAP+1.0ppm NAA; T_4: MS+6.0 ppm BAP+1.0ppm NAA; T_5: MS+7.0 ppm BAP+1.0ppm NAA). (Data within parenthesis are ArcSign transformed value. Means for groups in homogeneous subsets are displayed with same alphabets i.e. treatments denoted with same alphabets are not significant at 5% probability using DMRT. $CD_{0.05}$: 4.86; Error bar shows Standard deviation)

concerned with cell division, modification of apical dominance and shoot bud differentiation. In tissue culture media BAP is incorporated mainly for cell division and differentiation of shoot bud from organ or tissue (Bhojwani and Razdan, 1996). The role of cytokinin in in vitro establishment and multilplication of onion explant is also been reported (Passi *et al.*, 2018).

Number of shoots varied in different treatments with maximum (7.0) in MS+5.0 ppm BAP+1.0ppm KIN+0.5ppm GA3 where shoot length was also maximum (6.5 cm) (Fig. 2). No shoot bud was developed in medium containing 1.0 or 2.0 ppm BAP (T_1 and T_2). Addition of GA3 increased number of shoots. This may be due to elongation of small buds, which were not able to elongate without GA3. Increase in BAP concentration from 5.0 mg/l onward decreased the number of shoots. It was also observed that with the increase in the cytokinin concentration the shoot elongation was not affected as it was induced by the increase in GA3 level. Gibberellins are known to increase shoot length by inducing cell elongation and division (Rizza et al., 2017; Camara *et al.*, 2018; Suarez Padrón *et al.*, 2020).

The number of shoots varied with level of sucrose added to medium. With the increase in sucrose level from 30 to 40 g/l, number of shoots/cultures increased significantly but decreased thereafter. The maximum number of shoots (7.5) was recorded in medium containing 40 g/l of sucrose and minimum number of shoots (6.0) was found in treatment containing 60 g/l of sucrose. In tissue culture experiments, generally 2-4% sucrose (w/v) is usually optimum (George, 1993). Varying sucrose level have often given either increase or decrease in shoot multiplication rate and accordingly, it can be seen that the *in vitro* requirement of sucrose varies greatly with morphogenetic stage (Damino *et al.*, 1987).

The maximum number of shoots (7.0) were recorded at 5000 lux light intensity level and minimum (4.0) at 3000 lux. Shoot proliferation is generally influenced by light intensity. The optimum light intensity varies with plant species. Different morphogenetic processes require specific light intensity and both axillary and adventitious shoot bud proliferation are higher with increased light intensity (Lazzeri and Dunwell, 1986). There was difference between treatments with regard to number of shoots.

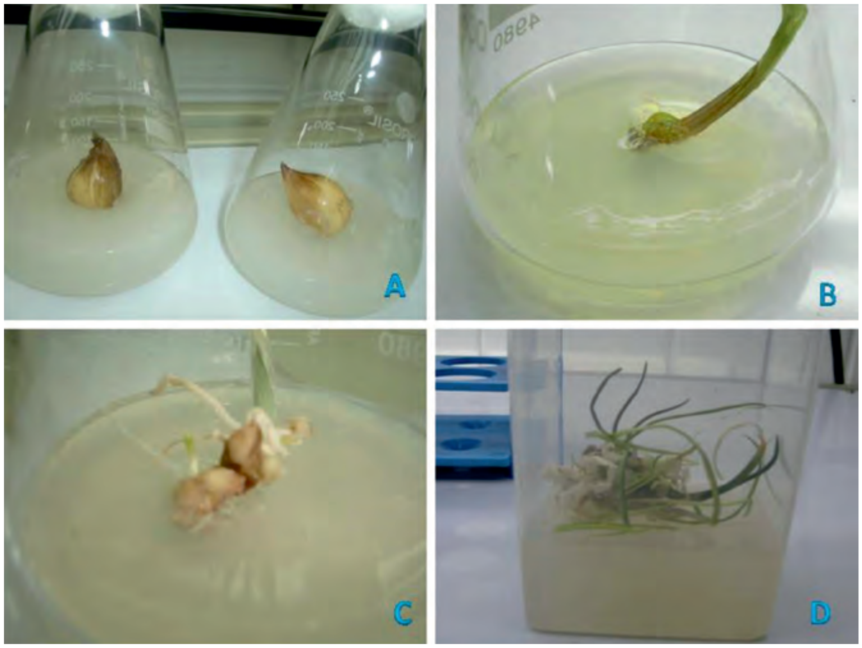

Fig. 2. A: *In vitro* culture establishment of garlic clove explant after treatment with s 0.1 % $HgCl_2$ for 2.0 minutes. B: sprouting of garlic clove on MS medium containing 2 mg/l GA3 and cloves were soaked in 100 mg/l GA3 solution for 24 hr. C: *In vitro* shoot bud induction on MS+4.0 ppm BAP+1.0ppm NAA. D: *In vitro* shoot multiplication on MS+5.0 ppm BAP+1.0ppm KIN+0.5ppm GA_3

Maximum number of shoots (7.0) were recorded when cultures were provided with 16:8 hr light and dark cycle and minimum (2.5) in 24:0 hr of light and dark cycle.

With increase or decrease in light period, number of shoots decreased. Shoot morphogenesis is generally stimulated by light and photoperiod coupled with total irradiance which has profound effect. In some plants, shoot proliferation is enhanced by high irradiation during stage-II (Hammerschag, 1978). Earlier, Haramaki (1971) suggested the use of light maintained at 3000-1000 lux for higher shoot proliferation in *Sinniongia*. Optimum light is also known to improve the quality of microshoots (Mcgranachau *et al*., 1987).

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